

Companion to Bird & de Moor, Algebra of Programming

§8. Thinning Algorithms

§8.1. Thinning

(8.1)

For $Q : A \multimap A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : PA \multimap PA$ (8.1).

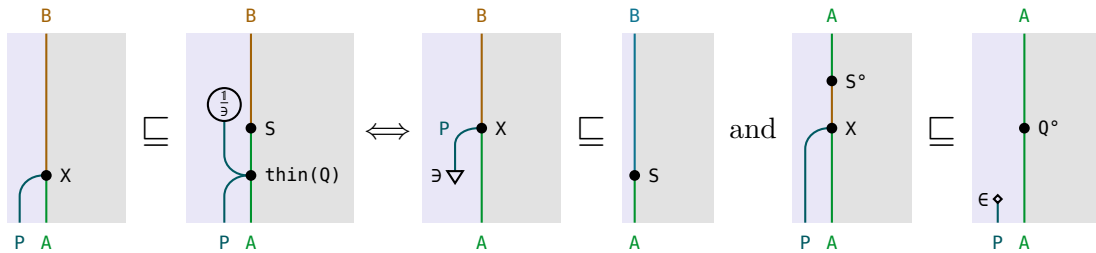
$ys \text{ thin}(Q) \ xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: ϵX runs $a \multimap ys \multimap xs$, member of ys first, and $Q^\circ \epsilon$ runs $a \multimap b \multimap xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q^\circ \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(8.1a)

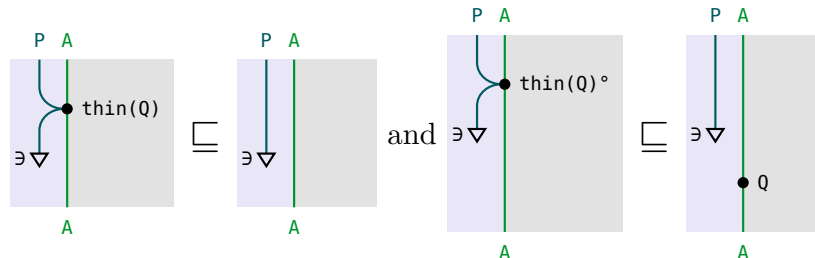
the law	what it says
1 $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \epsilon$	S is the whole algebra: it produces every candidate that $\text{thin}(Q)$ thins. Everything kept is an S -value, and every S -value has a Q -lower bound among the kept ones
2 $\text{thin}(Q) \ni \sqsubseteq \exists$	everything a thinning keeps was in the set
3 $\text{thin}(Q)^\circ \ni \sqsubseteq \exists \ Q$	every element of the set has a Q -lower bound among the kept ones
4 $Q \sqsubseteq R \implies \text{thin}(Q) \sqsubseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
5 $1 \sqsubseteq Q \implies 1 \sqsubseteq \text{thin}(Q)$ $Q \ Q \sqsubseteq Q \implies \text{thin}(Q) \ \text{thin}(Q) \sqsubseteq \text{thin}(Q)$	keeping everything is always a legal thinning
6 $Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge R \ R \sqsubseteq R \implies \text{est}(R) = \text{thin}(Q) \ \text{est}(R)$ $1 \sqsubseteq Q$ $\implies \text{est}(R) \sqsubseteq \text{thin}(Q) \ \text{est}(R)$	thin-introduction : thinning first cannot lose an R -minimum
7 $\text{thin}(Q) \sqsupseteq \text{est}(Q) \ \frac{1}{\exists} \quad (8.2) - (8.2) \ \text{est}(Q) \ \frac{1}{\exists} \sqsubseteq \text{thin}(Q)$	thin-elimination : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
8 $\frac{S}{\exists} \text{thin}(Q) \sqsupseteq \frac{S}{\exists} \text{est}(R) \ \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \sqsubseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
9 $P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q) \quad (8.4) - (8.4)$ $P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q)$	thinning each member set is a thinning of the union

§8.1.1. $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \epsilon$ — the definition of $\frac{S}{\exists} \text{thin}(Q)$

(8.1.1a)



(8.1.1b)

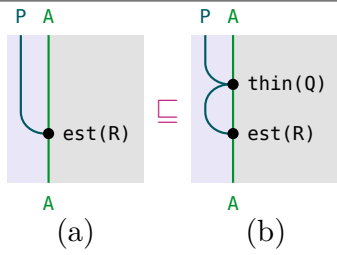


$X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \epsilon$ at $X \triangleq \text{thin}(Q)$, $S \triangleq \exists$: the left side is $\text{thin}(Q) \sqsubseteq \text{thin}(Q)$, since $\frac{\exists}{\exists} = \wedge(\exists) = 1$

§8.1.2. $Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge R \ R \sqsubseteq R \implies \text{est}(R) = \text{thin}(Q) \ \text{est}(R)$ — thinning first, then taking the smallest, is the same as taking the smallest

$$1 \sqsubseteq Q \Rightarrow \text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R) \quad (8.1.2a)$$

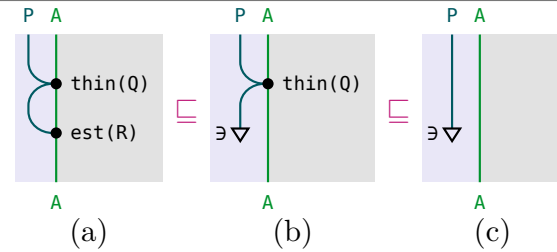
the $\sqsubseteq$ half: keeping everything is a thinning — $1 \sqsubseteq Q$



(a)	$\text{est}(R)$	$1 \sqsubseteq \text{thin}(Q) \text{ — } 1 \sqsubseteq Q$ $\Rightarrow 1 \sqsubseteq \text{thin}(Q)$, $Q \sqsubseteq Q$ $\Rightarrow \text{thin}(Q)$ $\text{thin}(Q) \sqsubseteq \text{thin}(Q)$
(b)	$\sqsubseteq \text{thin}(Q) \text{ est}(R)$	

$$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists \quad (8.1.2b)$$

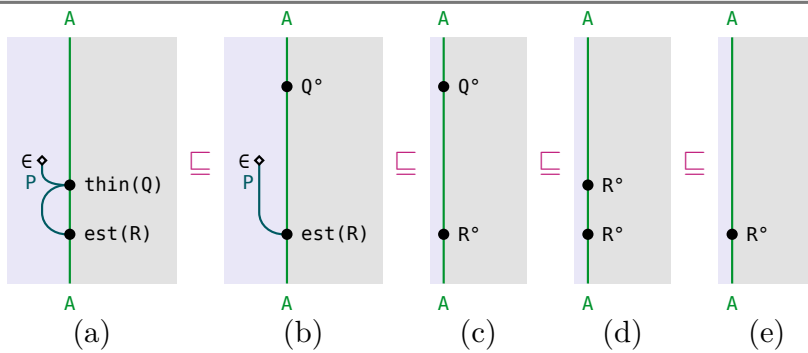
the $\sqsubseteq$ half, first condition of the UP of est at $X \triangleq \text{thin}(Q) \text{ est}(R) \text{ — } X \sqsubseteq \text{est}(R) \Leftrightarrow X \sqsubseteq \exists \wedge \in X \sqsubseteq R^\circ$



(a)	$\text{thin}(Q) \text{ est}(R)$	$R \cap S \sqsubseteq R$
(b)	$\sqsubseteq \text{thin}(Q) \exists$	$\text{thin}(Q) \exists \sqsubseteq \exists$
(c)	$\sqsubseteq \exists$	

$$Q \sqsubseteq R \wedge R R \sqsubseteq R \Rightarrow \in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ \quad (8.1.2c)$$

the $\sqsubseteq$ half, second condition



(a)	$\in \text{thin}(Q) \text{ est}(R)$	$\in \text{thin}(Q) \sqsubseteq Q^\circ \in$
(b)	$\sqsubseteq Q^\circ \in \text{est}(R)$	$\in \text{est}(R) \sqsubseteq R^\circ$
(c)	$\sqsubseteq Q^\circ R^\circ$	$Q \sqsubseteq R$
(d)	$\sqsubseteq R^\circ R^\circ$	$R R \sqsubseteq R$ $\Rightarrow R^\circ R^\circ \sqsubseteq R^\circ$
(e)	$\sqsubseteq R^\circ$	

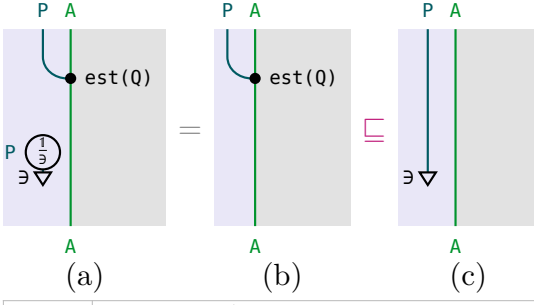
$$\mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq \mathbf{thin}(Q)$$

(8.2)

the singleton holding a Q -least member of a set is a thinning of that set

$$\mathbf{est}(Q) \stackrel{1}{\exists} \ni \sqsubseteq \ni$$

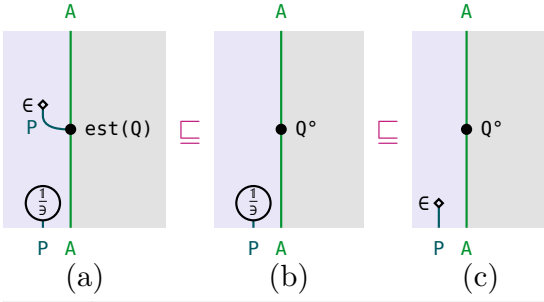
every member of the singleton is a member of the set — $\triangleq \mathbf{thin}, \ni/\ni$ half



(a)	$\mathbf{est}(Q) \stackrel{1}{\exists} \ni$	$\stackrel{1}{\exists} \ni = 1$
(b)	$= \mathbf{est}(Q)$	$R \cap S \sqsubseteq R$
(c)	$\sqsubseteq \ni$	

$$\in \mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \in$$

the singleton's member stands in Q to every member of the set — $\triangleq \mathbf{thin}, \in/(Q^\circ \in)$ half

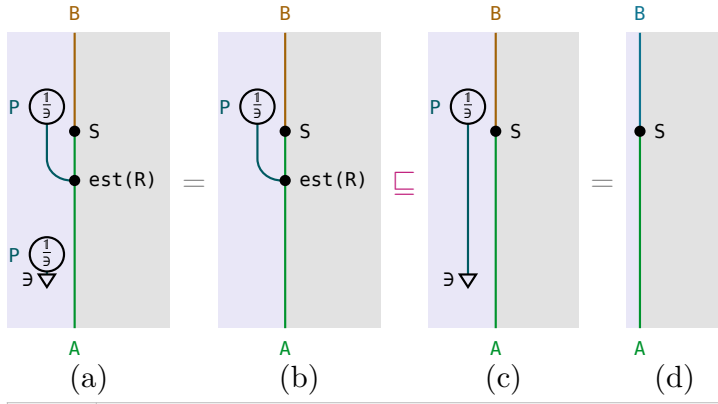


(a)	$\in \mathbf{est}(Q) \stackrel{1}{\exists}$	$\in \mathbf{est}(R) \sqsubseteq R^\circ$
(b)	$\sqsubseteq Q^\circ \stackrel{1}{\exists}$	$\stackrel{1}{\exists} \sqsubseteq \in$
(c)	$\sqsubseteq Q^\circ \in$	

$$R \cap (S^\circ S) \subseteq Q \Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq \frac{S}{\exists} \text{thin}(Q) \quad (8.3)$$

$$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni \in S$$

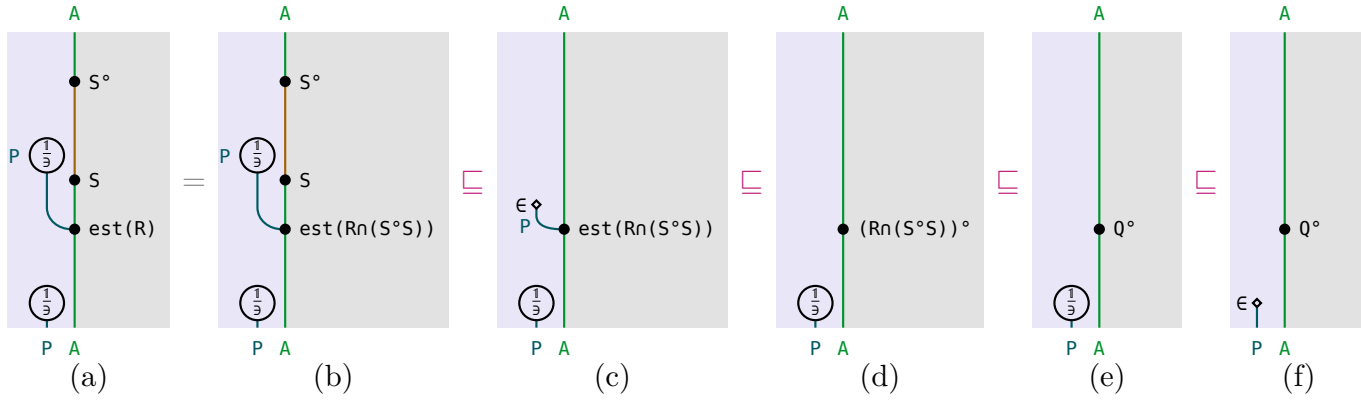
every member of the singleton is a value S returns — $X \subseteq \frac{S}{\exists} \text{thin}(Q) \Leftrightarrow X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$, first condition



(a)	$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni$	$\frac{1}{\exists} \ni = 1$
(b)	$= \frac{S}{\exists} \text{est}(R)$	$R \cap S \subseteq R$
(c)	$\subseteq \frac{S}{\exists} \ni$	$\frac{R}{\exists} \ni = R$
(d)	$= S$	

$$R \cap (S^\circ S) \subseteq Q \Rightarrow S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq Q^\circ \in$$

the singleton's member stands in Q to every value S returns at the same argument — $X \subseteq \frac{S}{\exists} \text{thin}(Q) \Leftrightarrow X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$, second condition



(a)	$S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$	$\frac{S}{\exists} \text{est}(R \cap (S^\circ S)) = \frac{S}{\exists} \text{est}(R)$
(b)	$= S^\circ \frac{S}{\exists} \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$	$T^\circ \frac{1}{\exists} \subseteq \in$
(c)	$\subseteq \in \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$	$\in \text{est}(R) \subseteq R^\circ$
(d)	$\subseteq (R \cap (S^\circ S))^\circ \frac{1}{\exists}$	$R \cap (S^\circ S) \subseteq Q$
(e)	$\subseteq Q^\circ \frac{1}{\exists}$	$\frac{1}{\exists} \subseteq \in$
(f)	$\subseteq Q^\circ \in$	

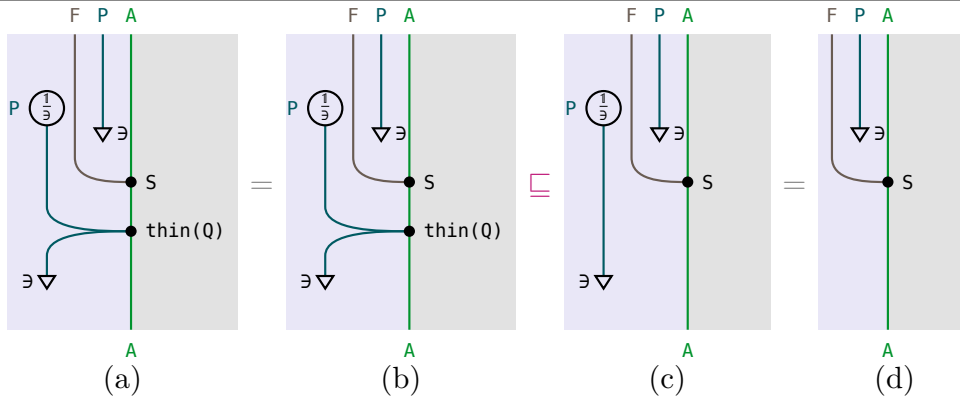
$$\text{preorder}(Q) \wedge F(Q) S \sqsubseteq S Q \implies (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq (\frac{S}{\exists} \text{thin}(Q))$$

Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set

$$(\frac{F(\exists) S}{\exists} \text{thin}(Q)) \ni \sqsubseteq (S)$$

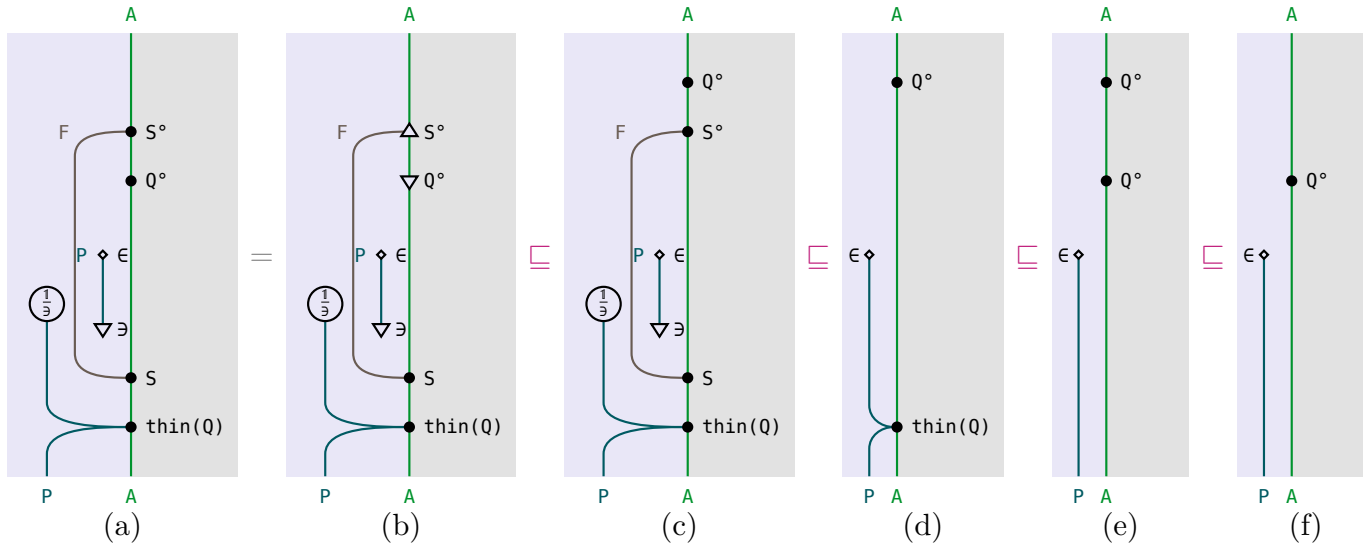
everything the thinning fold keeps is a value of the plain fold — $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, first condition, by fusion from the chain below



(a)	$\frac{F(\exists) S}{\exists} \text{thin}(Q) \ni$	
(b)	$= \frac{F(\exists) S}{\exists} \text{thin}(Q) \ni$	$\text{thin}(Q) \ni \sqsubseteq \exists$
(c)	$\sqsubseteq \frac{F(\exists) S}{\exists} \ni$	$\frac{R}{\exists} \ni = R$
(d)	$= F(\exists) S$	

$$\text{preorder}(Q) \wedge F(Q) S \sqsubseteq S Q \implies (S)^\circ (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq Q^\circ \in$$

every value of the plain fold is Q -above one the thinning fold keeps — $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, second condition, by hylomorphism theorem from the chain below



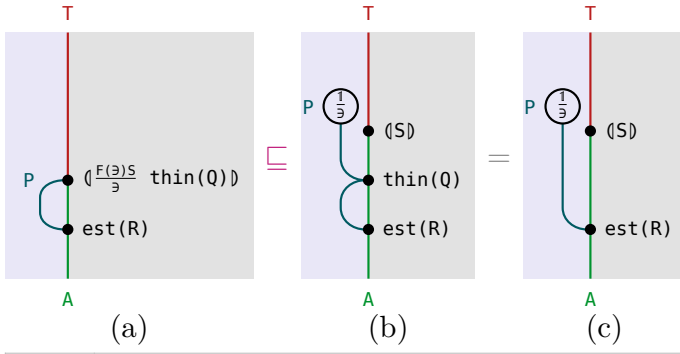
(a)	$S^\circ F(Q^\circ \in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	
(b)	$= S^\circ F(Q^\circ) F(\epsilon) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$F(Q) S \sqsubseteq S Q$
(c)	$\sqsubseteq Q^\circ S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$\implies S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$
(d)	$\sqsubseteq Q^\circ \in \text{thin}(Q)$	$S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \sqsubseteq \epsilon$
(e)	$\sqsubseteq Q^\circ Q^\circ \in$	$\epsilon \text{thin}(Q) \sqsubseteq Q^\circ \in$
(f)	$\sqsubseteq Q^\circ \in$	

(8.1.2d)

	step	type	example	what it does
1	$S:F(A)\rightarrow A$ is 8.1's algebra; example $F(X)=V+V\times X$, $S=[\text{wrap},\text{cons}]$, candidate $a=\text{cons}(v,p)$			
2	S°	$A\rightarrow FA$	$\text{cons}(v,p) \mapsto r(v,p)$	takes a apart into the input of its last step
3	$F(Q^\circ\in)$	$FA\rightarrow F(PA)$	$r(v,p) \mapsto r(v,ps)$, some $p'\in ps$ with $p' \ Q \ p$	replaces each recursive component by a set holding a Q -no-worse one; v stays
4	$\frac{F(\exists)S}{\exists}$	$F(PA)\rightarrow PA$	$r(v,ps) \mapsto \{\text{cons}(v,t) \mid t\in ps\}$	picks one element from each set, combines by S , collects every result
5	$\text{thin}(Q)$	$PA\rightarrow PA$	$\{\text{cons}(v,t) \mid t\in ps\} \mapsto ys$	keeps a subset; every dropped element is Q -beaten by a kept one
6	$\sqsubseteq Q^\circ\in$	$A\rightarrow PA$	ys holds y with $y \ Q \ \text{cons}(v,p') \ Q \ \text{cons}(v,p) = a$	the thinned set still holds an element Q -no-worse than a (S monotonic on Q , Q transitive)

$$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(Q) S \sqsubseteq S Q \implies (\downarrow \frac{F(\exists) S}{\exists} \text{thin}(Q)) \text{est}(R) \sqsubseteq \frac{(S)}{\exists} \text{est}(R) \quad \text{Corollary 8.1}$$

the thinning fold refines the optimisation problem itself



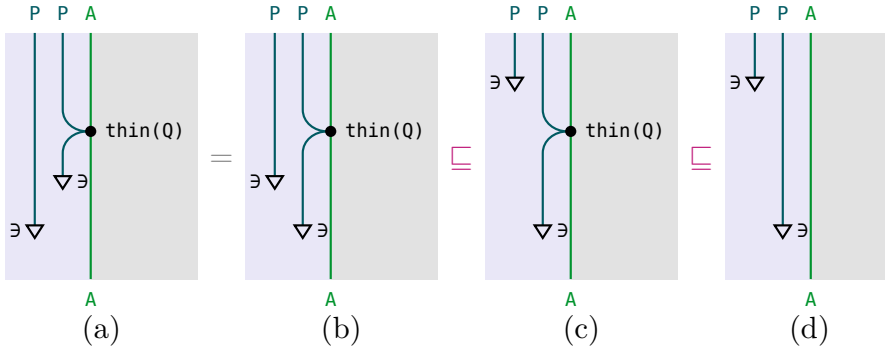
(a)	$(\downarrow \frac{F(\exists) S}{\exists} \text{thin}(Q)) \text{est}(R)$	Theorem 8.1 $\text{preorder}(Q) \wedge F(Q) S \sqsubseteq S Q$ $\implies (\downarrow \frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq \frac{(S)}{\exists} \text{thin}(Q)$
(b)	$\sqsubseteq \frac{(S)}{\exists} \text{thin}(Q) \text{est}(R)$	
(c)	$= \frac{(S)}{\exists} \text{est}(R)$	

$$P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q) \quad (8.4)$$

thinning each member set, then taking the union, is a thinning of the union

$$P(\text{thin}(Q)) \text{ union} \exists \sqsubseteq \exists \exists$$

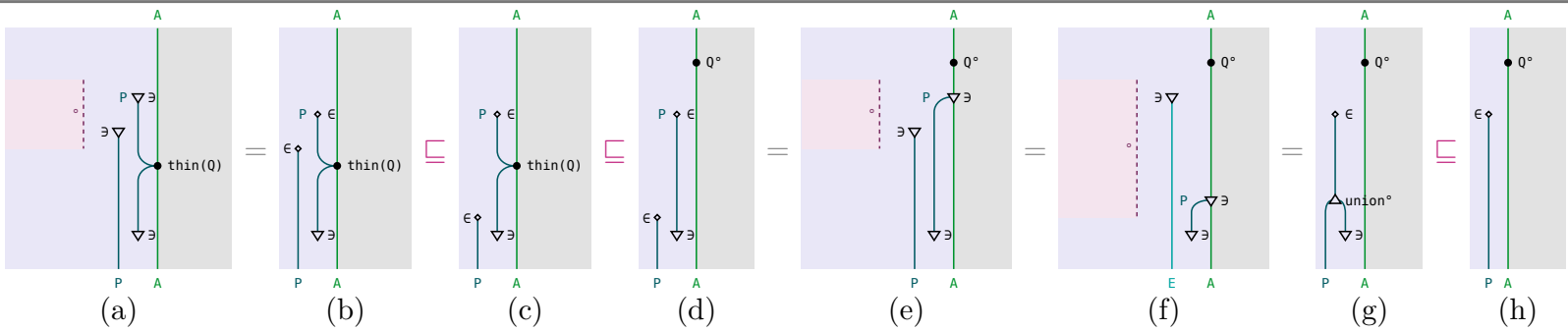
every member of the result is a member of a member set — $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, first condition



(a)	$P(\text{thin}(Q)) \text{ union} \exists$	$\text{union} \exists = \exists \exists$
(b)	$= P(\text{thin}(Q)) \exists \exists$	$P(R) \exists \sqsubseteq \exists R$
(c)	$\sqsubseteq \exists \text{thin}(Q) \exists$	$\text{thin}(Q) \exists \sqsubseteq \exists$
(d)	$\sqsubseteq \exists \exists$	

$$(\exists \exists)^\circ P(\text{thin}(Q)) \text{ union} \sqsubseteq Q^\circ \in$$

every member of a member set has a Q -lower bound in the result — $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, second condition



(a)	$(\exists \exists)^\circ P(\text{thin}(Q)) \text{ union}$	$(R S)^\circ = S^\circ R^\circ$
(b)	$= \in \in P(\text{thin}(Q)) \text{ union}$	$\in P(R) \sqsubseteq R \in$
(c)	$\sqsubseteq \in \text{thin}(Q) \in \text{union}$	$\in \text{thin}(Q) \sqsubseteq Q^\circ \in$
(d)	$\sqsubseteq Q^\circ \in \in \text{union}$	$(R S)^\circ = S^\circ R^\circ$
(e)	$= Q^\circ (\exists \exists)^\circ \text{union}$	$\text{union} \exists = \exists \exists$
(f)	$= Q^\circ (\text{union} \exists)^\circ \text{union}$	$(R S)^\circ = S^\circ R^\circ$
(g)	$= Q^\circ \in \text{union}^\circ \text{union}$	$\text{union} = E(\exists)$
(h)	$\sqsubseteq Q^\circ \in$	

§8.2. Paths in a layered network

(8.2a)

	name	type	definition	meaning
1	F	$\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$	$F(B, X) = B + (B \times X)$	a path is either one vertex a , or a vertex a followed by a path xs
2	α	$F(A, LA) \rightarrow LA$	$\alpha = [\text{wrap}, \text{cons}]$	build a path: wrap one vertex as $[a]$, or put a in front of xs
3	wrapz	$A \rightarrow LA \times \mathbb{N}$	$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle, a \mapsto ([a], 0)$	a path of one vertex, no edge, cost 0
4	consw	$A \times (LA \times \mathbb{N}) \rightarrow LA \times \mathbb{N}$	$(a, (xs, n)) \mapsto ([a] \# xs, \text{wt}(a, \text{head}(xs)) + n)$	put a in front of the path; the cost adds the weight of the new edge $a \rightarrow \text{head}(xs)$
5	cost	$LA \rightarrow \mathbb{N}$	$\text{cost} \triangleq ([\text{wrapz}, \text{consw}]) \circ \pi_2$	the cost of $[a_0, \dots, a_n]$ is $\text{wt}(a_0, a_1) + \dots + \text{wt}(a_{n-1}, a_n)$
6	R	$LA \rightarrow LA$	$R \triangleq \text{cost} \leq \text{cost}^\circ$	xs is cheaper than ys
7	$L(R)$	$LA \rightarrow LB$	$([F(\exists, 1) \alpha]) = L(\exists)$	$L(\exists)$ relates a list of layers to every path that chooses one vertex from each layer
8	minpath	$L(PA) \rightarrow LA$	$\text{minpath} \sqsubseteq \frac{L(\exists)}{\exists} \text{est}(R)$	the input $[x_0, \dots, x_n] : L(PA)$ is a list of layers, each a set of vertices; a path chooses one vertex from each layer, and minpath returns a cheapest one
9	Q	$LA \rightarrow LA$	$Q \triangleq R \cap (\text{head} \text{ head}^\circ)$	like R , and the two paths also start at the same vertex: $\text{head}(xs) = \text{head}(ys)$
10	l	$B \rightarrow B + C$	$a \mapsto l(a)$	mark a value as the left alternative
11	r	$C \rightarrow B + C$	$b \mapsto r(b)$	mark a value as the right alternative
12	cpl	$PB \times A \rightarrow P(B \times A)$	$\text{cpl} \triangleq \text{cp}(1 \times A)$	pair every element of the set with the value beside it
13	cpr	$A \times PB \rightarrow P(A \times B)$	$\text{cpr} \triangleq \text{cp}(A \times 1)$	pair the value beside the set with every element of the set
14	$\frac{F(\exists, 1)}{\exists}$	$F(PB, X) \rightarrow PF(B, X)$	$\frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$	cpl transpose: from layer x and a set of paths ps , choose a vertex $a \in x$
15	$\frac{F(1, \exists)}{\exists}$	$F(B, PX) \rightarrow PF(B, X)$	$\frac{F(1, \exists)}{\exists} = [\frac{1}{\exists} P(l), \text{cpr } P(r)]$	cpr transpose: from a vertex a and a set of paths ps , choose a tail $xs \in ps$
16	step	$A \times P(LA) \rightarrow LA$	$\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$	put a in front of every path in ps and keep a cheapest one

(8.2b)

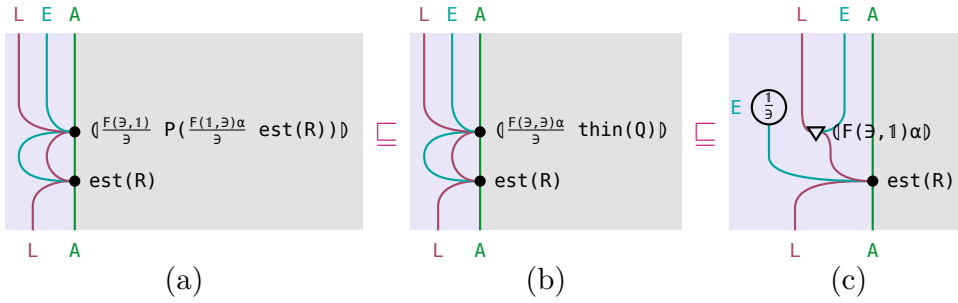
	the law	what it says
1	$F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
2	$S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	$S \text{ head}$ is simple, which gives $R \cap (S^\circ S) \sqsubseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

(8.2a)

$$\left(\frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \text{ est}(R)\right) \right) \text{ est}(R) \sqsubseteq \left(\frac{F(\exists,1)}{\exists} \alpha \right) \text{ est}(R) \quad (8.2c)$$

a least-cost path in a layered network, as a fold over the layers

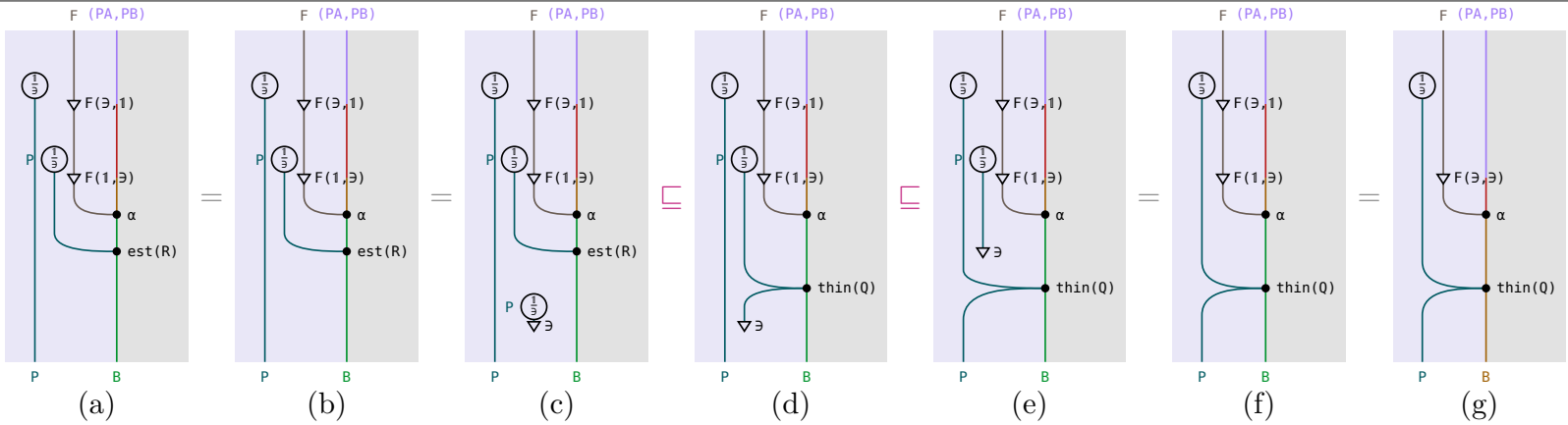
- Why it is needed: $\text{est}(R)$ keeps one cheapest w of the set $\Lambda(S)(x)$; $\text{thin}(Q)$ needs every dropped z to be Q -beaten by a kept one. For $\{w\}$ alone to be a thinning we need $w \leq Q \leq z$ for every z in $\Lambda(S)(x)$.
- We have $w \leq R \leq z$ (w is cheapest) and $w \leq (S^\circ S) \leq z$ (both are built by S from the same argument x); the hypothesis turns the two into $w \leq Q \leq z$: among candidates from one argument, cheaper already means Q -better.



(a)	$\left(\frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \text{ est}(R)\right) \right) \text{ est}(R)$	$\frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \text{ est}(R)\right) \sqsubseteq \frac{F(\exists,\exists)}{\exists} \text{ thin}(Q)$
(b)	$\sqsubseteq \left(\frac{F(\exists,\exists)}{\exists} \alpha \text{ thin}(Q) \right) \text{ est}(R)$	$\left(\frac{F(\exists,\exists)}{\exists} \alpha \text{ thin}(Q) \right) \text{ est}(R) \sqsubseteq \left(\frac{F(\exists,1)}{\exists} \alpha \right) \text{ est}(R)$
(c)	$\sqsubseteq \left(\frac{F(\exists,1)}{\exists} \alpha \right) \text{ est}(R)$	

$$S \triangleq F(1,\exists) \alpha, R \sqcap (S^\circ S) \sqsubseteq Q \Rightarrow \frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \text{ est}(R)\right) \sqsubseteq \frac{F(\exists,\exists)}{\exists} \alpha \text{ thin}(Q) \quad (8.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases



(a)	$\frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \text{ est}(R)\right)$	
(b)	$= \frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \text{ est}(R)\right)$	$P\left(\frac{1}{\exists}\right) E(\exists) = 1$
(c)	$= \frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \text{ est}(R) \frac{1}{\exists}\right) E(\exists)$	(8.3) $R \sqcap (S^\circ S) \sqsubseteq Q$ $\Rightarrow \frac{S}{\exists} \text{ est}(R) \frac{1}{\exists} \sqsubseteq \frac{S}{\exists} \text{ thin}(Q)$
(d)	$\sqsubseteq \frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \right) P(\text{thin}(Q)) E(\exists)$	(8.4) $P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q)$
(e)	$\sqsubseteq \frac{F(\exists,1)}{\exists} P\left(\frac{F(1,\exists)}{\exists} \alpha \right) E(\exists) \text{ thin}(Q)$	$\frac{V S}{\exists} = \frac{V}{\exists} P\left(\frac{S}{\exists}\right) \text{ union}$
(f)	$= \frac{F(\exists,1)}{\exists} \frac{F(1,\exists)}{\exists} \alpha \text{ thin}(Q)$	$F(R,1) F(1,S) = F(R,S)$
(g)	$= \frac{F(\exists,\exists)}{\exists} \alpha \text{ thin}(Q)$	

$$= \text{cp}(F) P(\alpha) \text{ est}(R) = [\text{wrap}, \text{step}]$$

at the layered network $(F(A,X)=A+A \times X, B \text{ the paths } V^+, \alpha=[\text{wrap}, \text{cons}]): \frac{F(1,\exists)}{\exists} P(\alpha) \text{ est}(R)=[\text{wrap}, \text{step}] - \text{step} \triangleq \text{cpr}$
 $P(\text{cons}) \text{ est}(R)$

$$= \frac{F(\exists,1)}{\exists} = 1 + \text{cpl} - \frac{F(\exists,1)}{\exists} = [P(l), \text{cpl } P(r)]$$

$$\begin{array}{ccccc} F(A,X) & = & A & + & A \times X \\ & & | & & | \times | \\ & & R & & R \quad S \\ & & \downarrow & & \downarrow \quad \downarrow \\ F(A',X') & = & A' & + & A' \times X' \end{array} \quad (8.2e)$$

u $F(R,S)$ v holds iff one of:

- u, v are both inl and $d \leq R \leq d'$: the leaf moves by R;
- u, v are both inr and $p_1 \leq R \leq q_1 \wedge p_2 \leq S \leq q_2$: the vertex moves by R, the rest by S.
- An inl never relates to an inr: $R+S \triangleq [R \text{ inl}, S \text{ inr}]$ returns to the summand it came from, and $F(1)=1$ allows no crossing.
- $F(\exists,\exists)$ relates $\text{inr}(\{a,b\},\{p,q\})$ to $\text{inr}(a,p), \text{inr}(a,q), \text{inr}(b,p), \text{inr}(b,q)$, and $\text{inl}(\{a,b\})$ to $\text{inl}(a), \text{inl}(b)$.

§8.3. Implementing thin

(8.3a)

	name	type	definition	meaning
1	setify	$[A] \rightarrow PA$	$(\text{setify } \exists)(x, a) \iff a \in x$	the set of a list's elements
2	cup	$PA \times PA \rightarrow PA$	$\text{cup} \triangleq \frac{(\pi_1 \exists) \cup (\pi_2 \exists)}{\exists}$	the union of two sets
3	cp(F)	$F(PA) \rightarrow PFA$	$\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$	every F-shaped value whose parts are chosen from the given sets
4	listcp(F, A, X)	$\text{relator}(\text{list}(X)) \rightarrow \text{list}(\text{relator}(X))$	$\text{listcpFn}(\text{zer}, x) = []$, $\text{listcpFn}(\text{PolyC.one}, u) = [u, ()]$, $\text{listcpFn}(\text{const}(a), c) = [c]$, $\text{listcpFn}(\text{arg}_1, a) = [a]$, $\text{listcpFn}(\text{arg}_2, xs) = xs$, $\text{listcpFn}(l \otimes a, l(u)) =$ $\text{list}(\text{inl}, \text{listcpFn}(l, u))$, $\text{listcpFn}(a \otimes r, r(v)) = \text{list}(\text{inr}, \text{listcpFn}(r, v))$, $\text{listcpFn}(l \otimes r, (u, v)) = \text{cppFn}(\text{listcpFn}(l, u), \text{listcpFn}(r, v))$	by the shape of F: a constant or arg_1 is a one-element list, arg_2 the list itself, a sum lists its summand, a product pairs every element of one list with every element of the other
5	linear	$P(\text{PolyC}(\imath))$	$\text{linear}(\text{zer}) = \text{true}$, $\text{linear}(\text{PolyC.one}) = \text{true}$, $\text{linear}(\text{const}(a)) = \text{true}$, $\text{linear}(\text{arg}_1) = \text{true}$, $\text{linear}(\text{arg}_2) = \text{true}$, $\text{linear}(l \otimes r) = \text{linear}(l) \wedge \text{linear}(r)$, $\text{linear}(l \otimes r) = \text{linear}(l) \wedge \text{linear}(r) \wedge (\neg \text{hasArg}_2(l) \vee \neg \text{hasArg}_2(r))$	no product in F has the argument on both sides
6	sort($\leq$)	$PA \rightarrow LA$	$\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ordered}(\leq)$	read a set back as one of its $\leq$ -ordered listings
7	bump(Q)	$A \times [A] \rightarrow [A]$	$(a, []) \mapsto [a]$, $\text{bump}(Q)((a, [b] \# xs), ys) \iff$ $Q(a, b) \wedge ys = [a] \# xs \vee Q(b, a) \wedge ys = [b] \# xs$ $\vee \neg Q(a, b) \wedge \neg Q(b, a) \wedge ys = [a] \# [b] \# xs$	put a in front of a thinned list, dropping whichever of a and the old head the other beats
8	thinlist(Q)	$[A] \rightarrow [A]$	$\text{thinlist}(Q) = ([\text{nil}, \text{bump}(Q)])^\flat$	thin a list in one pass, bumping each element into the thinned rest
9			$\text{IsThinlist}(Q, \text{thinlist}) \iff \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{thinlist setify} \sqsubseteq \text{setify thin}(Q)$	what a list implementation of $\text{thin}(Q)$ must do: only drop elements, and thin the set
10	minlist(Q)	$[A] \rightarrow A$	$\text{minlist}(Q) \triangleq \text{setify est}(Q)$	a Q-least element of the list

(8.3b)

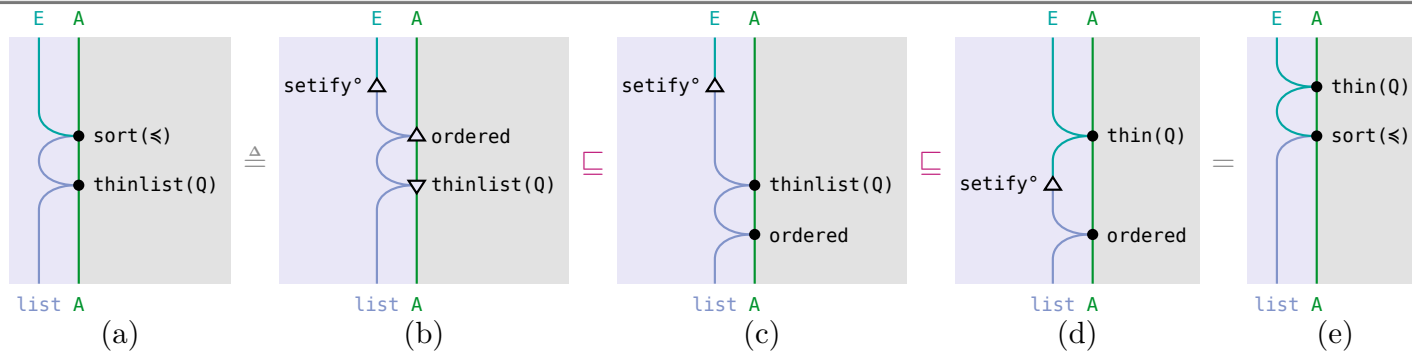
Binary thinning data: Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; $\leq$ a connected preorder with both f_1, f_2 monotonic on $\leq$; R a preorder.

(8.3c)

the law	what it says
1 $\text{preorder}(Q) \wedge \text{connected}(Q) \Rightarrow \text{thinlist}(Q)$ $([a]\#xs,ys) \Leftrightarrow (\exists w. \text{minlist}(Q)([a]\#xs,w) \wedge ys = [w])$ (8.5)	what thinning should come to when it can: one element
2 $\text{preorder}(Q) \Rightarrow \text{sort}(\leq) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q)$ $\text{sort}(\leq)$ (8.6) — (8.6) $\text{preorder}(Q)$ $\Rightarrow \text{sort}(\leq) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(\leq)$	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
3 $\text{sort}(\leq) \text{minlist}(R) \sqsubseteq \text{est}(R)$ (8.7)	a minimum of the sorted list is a minimum of the set
4 $\text{sort}(g \leq g^\circ) \text{list}(g) \sqsubseteq P(g) \text{sort}(\leq)$ (8.8)	shunt a function through a sort
5 $\text{sort}(\leq) \text{filter}(p) \sqsubseteq E(p) \text{sort}(\leq)$ (8.9)	filtering a sorted list sorts the restricted set
6 $\text{preorder}(\leq) \wedge \text{connected}(\leq) \Rightarrow (\text{sort}(\leq) \times \text{sort}(\leq))$ $\text{merge}(\leq) \sqsubseteq \text{cup sort}(\leq)$ (8.10)	merging two sorted lists sorts their union
7 $F(\text{sort}(\leq)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sort}(F(\leq))$ (8.11), $FX=L+E \times X$	listcp is the list implementation of the cartesian product $\text{cp}(F)$
8 $\text{linear}(F) \Rightarrow \text{relator}(\text{sort}(\leq)) \text{listcp}(F,A,X) \sqsubseteq$ $\text{cp}(\text{relator}(F,A)) \text{sort}(\text{relator}(\leq))$ (8.11), F polynomial and linear	the same for every linear polynomial F

$$\text{preorder}(Q) \Rightarrow \text{sort}(\leq) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(\leq) \quad (8.6)$$

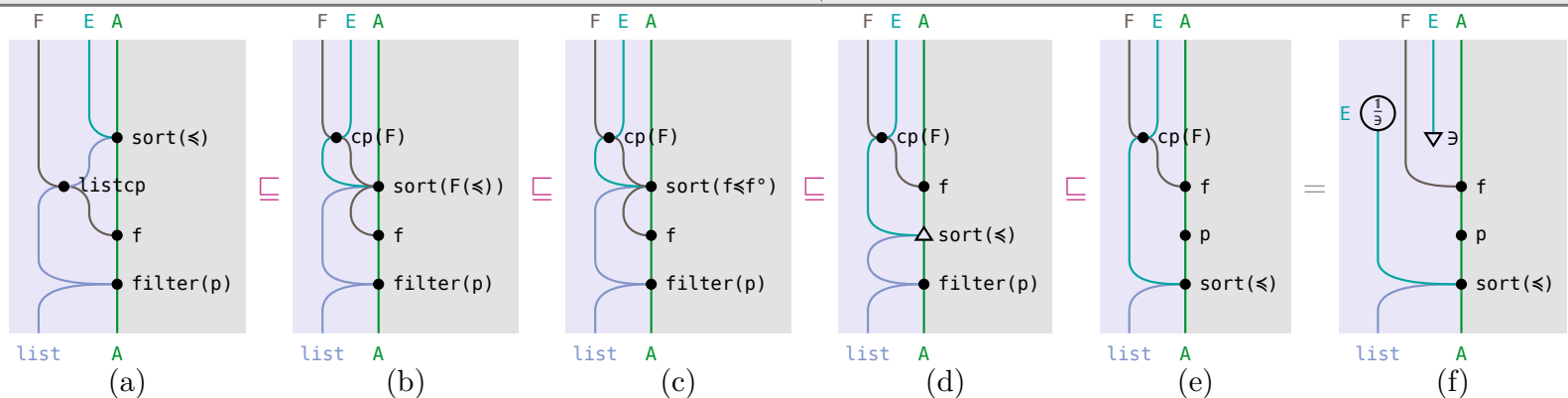
sorting the set then thinning the list by Q lists, sorted, a Q -thinning of the set — Q a preorder.



(a)	$\text{sort}(\leq) \text{thinlist}(Q)$	$\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ordered}(\leq)$
(b)	$\triangleq \text{setify}^\circ \text{ordered thinlist}(Q)$	$\text{ordered thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ordered}$
(c)	$\sqsubseteq \text{setify}^\circ \text{thinlist}(Q) \text{ordered}$	$\text{preorder}(Q)$
(d)	$\sqsubseteq \text{thin}(Q) \text{setify}^\circ \text{ordered}$	$\Rightarrow \text{setify}^\circ \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{setify}^\circ$
(e)	$= \text{thin}(Q) \text{sort}(\leq)$	

$$F(\leq) f \sqsubseteq f \leq \Rightarrow F(\text{sort}(\leq)) \text{listcp list}(f) \text{filter}(p) \sqsubseteq \frac{F(\exists) f p}{\exists} \text{sort}(\leq) \quad \text{Lemma 8.1}$$

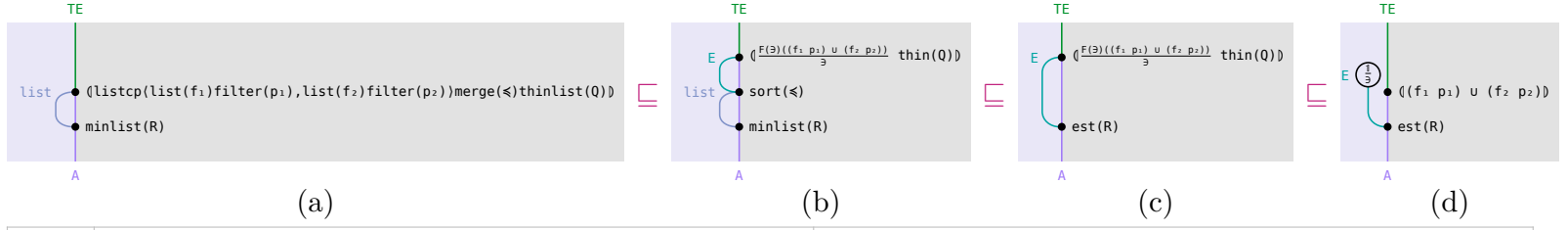
one sorted list built from sorted arguments, instead of a set built and then sorted — $f : FA \rightarrow A$ monotonic on $\leq$.



(a)	$F(\text{sort}(\leq)) \text{listcp list}(f) \text{filter}(p)$	$F(\text{sort}(\leq)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sort}(F(\leq))$
(b)	$\sqsubseteq \text{cp}(F) \text{sort}(F(\leq)) \text{list}(f) \text{filter}(p)$	$F(\leq) f \sqsubseteq f \leq$
(c)	$\sqsubseteq \text{cp}(F) \text{sort}(f \leq f^\circ) \text{list}(f) \text{filter}(p)$	$\text{sort}(g \leq g^\circ) \text{list}(g) \sqsubseteq P(g) \text{sort}(\leq)$
(d)	$\sqsubseteq \text{cp}(F) P(f) \text{sort}(\leq) \text{filter}(p)$	$\text{sort}(\leq) \text{filter}(p) \sqsubseteq E(p) \text{sort}(\leq)$
(e)	$\sqsubseteq \text{cp}(F) E(f p) \text{sort}(\leq)$	$\frac{S}{\exists} E(R) = \frac{S R}{\exists}$
(f)	$= \frac{F(\exists) f p}{\exists} \text{sort}(\leq)$	

Theorem 8.2
 $Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(Q) \ f_1 \ p_1 \sqsubseteq f_1 \ p_1 \ Q \wedge F(Q) \ f_2 \ p_2 \sqsubseteq f_2 \ p_2 \ Q \wedge$
 $\text{preorder}(\leq) \wedge \text{connected}(\leq) \wedge F(\leq) \ f_1 \sqsubseteq f_1 \leq \wedge F(\leq) \ f_2 \sqsubseteq f_2 \leq \implies (\text{listcp}(\text{list}(f_1) \text{ filter}(p_1), \text{list}(f_2) \text{ filter}(p_2)) \text{ merge}(\leq) \text{ thinlist}(Q)) \text{ minlist}(R) \sqsubseteq \frac{((f_1 \ p_1) \cup (f_2 \ p_2))}{\exists} \text{ est}(R)$

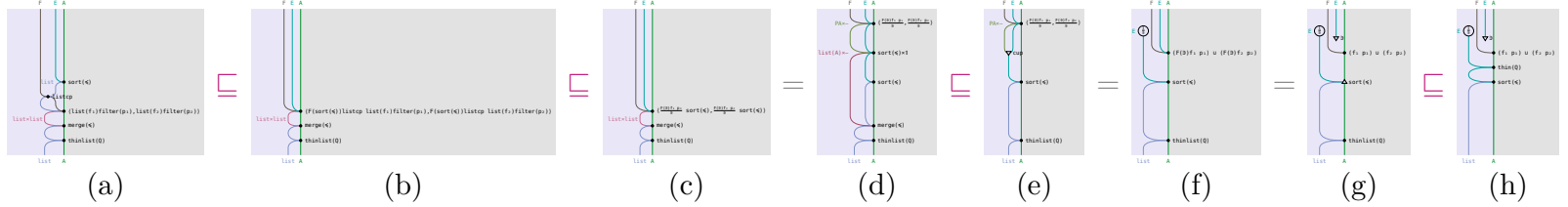
a fold on sorted lists of partial solutions, thinned at every step, refines the thinning specification — at binary thinning data.



(a)	$\text{listcp}(\text{list}(f_1) \text{ filter}(p_1), \text{list}(f_2) \text{ filter}(p_2)) \text{ merge}(\leq) \text{ thinlist}(Q) \text{ minlist}(R)$	(8.3d) $\text{preorder}(Q) \wedge \text{preorder}(\leq) \wedge \text{connected}(\leq) \wedge F(\leq) \ f_1 \sqsubseteq f_1 \leq \wedge F(\leq) \ f_2 \sqsubseteq f_2 \leq \implies F(\text{sort}(\leq)) \text{ listcp}(\text{list}(f_1) \text{ filter}(p_1), \text{list}(f_2) \text{ filter}(p_2)) \text{ merge}(\leq) \text{ thinlist}(Q) \sqsubseteq \frac{F(\exists) ((f_1 \ p_1) \cup (f_2 \ p_2))}{\exists} \text{ thin}(Q) \text{ sort}(\leq)$
(b)	$\sqsubseteq \frac{F(\exists) ((f_1 \ p_1) \cup (f_2 \ p_2))}{\exists} \text{ thin}(Q) \text{ sort}(\leq) \text{ minlist}(R)$	$\text{sort}(\leq) \text{ minlist}(R) \sqsubseteq \text{est}(R)$
(c)	$\sqsubseteq \frac{F(\exists) ((f_1 \ p_1) \cup (f_2 \ p_2))}{\exists} \text{ thin}(Q) \text{ est}(R)$	$F(Q) \ S_1 \sqsubseteq S_1 \ Q \wedge F(Q) \ S_2 \sqsubseteq S_2 \ Q \implies F(Q) \ (S_1 \cup S_2) \sqsubseteq (S_1 \cup S_2) \ Q$
(d)	$\sqsubseteq \frac{((f_1 \ p_1) \cup (f_2 \ p_2))}{\exists} \text{ est}(R)$	

(8.3d)
 $\text{preorder}(Q) \wedge \text{preorder}(\leq) \wedge \text{connected}(\leq) \wedge F(\leq) \ f_1 \sqsubseteq f_1 \leq \wedge F(\leq) \ f_2 \sqsubseteq f_2 \leq \implies F(\text{sort}(\leq))$
 $\text{listcp}(\text{list}(f_1) \text{ filter}(p_1), \text{list}(f_2) \text{ filter}(p_2)) \text{ merge}(\leq) \text{ thinlist}(Q) \sqsubseteq \frac{F(\exists) ((f_1 \ p_1) \cup (f_2 \ p_2))}{\exists} \text{ thin}(Q) \text{ sort}(\leq)$

sorting the candidate set turns the thinning algebra into an algebra on sorted lists.



(a)	$F(\text{sort}(\leq)) \text{ listcp}(\text{list}(f_1) \text{ filter}(p_1), \text{list}(f_2) \text{ filter}(p_2)) \text{ merge}(\leq) \text{ thinlist}(Q)$	$X \langle R, S \rangle \sqsubseteq \langle X \ R, X \ S \rangle$
(b)	$\sqsubseteq (F(\text{sort}(\leq)) \text{ listcp}(\text{list}(f_1) \text{ filter}(p_1), F(\text{sort}(\leq)) \text{ listcp}(\text{list}(f_2) \text{ filter}(p_2)) \text{ merge}(\leq) \text{ thinlist}(Q))$	Lemma 8.1 $F(\leq) \ f \sqsubseteq f \leq \implies F(\text{sort}(\leq)) \text{ listcp}(\text{list}(f) \text{ filter}(p)) \sqsubseteq \frac{F(\exists) \ f \ p}{\exists} \text{ sort}(\leq)$
(c)	$\sqsubseteq (\frac{F(\exists) \ f_1 \ p_1}{\exists} \text{ sort}(\leq), \frac{F(\exists) \ f_2 \ p_2}{\exists} \text{ sort}(\leq)) \text{ merge}(\leq) \text{ thinlist}(Q)$	$\langle X, Y \rangle (R \times S) = \langle X \ R, Y \ S \rangle$
(d)	$= (\frac{F(\exists) \ f_1 \ p_1}{\exists}, \frac{F(\exists) \ f_2 \ p_2}{\exists}) (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \text{ thinlist}(Q)$	$\text{preorder}(\leq) \wedge \text{connected}(\leq) \implies (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq)$
(e)	$\sqsubseteq (\frac{F(\exists) \ f_1 \ p_1}{\exists}, \frac{F(\exists) \ f_2 \ p_2}{\exists}) \text{ cup} \text{ sort}(\leq) \text{ thinlist}(Q)$	$\frac{R \cup S}{\exists} = \langle \frac{R}{\exists}, \frac{S}{\exists} \rangle \text{ cup}$
(f)	$= \frac{(F(\exists) \ f_1 \ p_1) \cup (F(\exists) \ f_2 \ p_2)}{\exists} \text{ sort}(\leq) \text{ thinlist}(Q)$	$R \ (S \cup T) = (R \ S) \cup (R \ T)$
(g)	$= \frac{F(\exists) ((f_1 \ p_1) \cup (f_2 \ p_2))}{\exists} \text{ sort}(\leq) \text{ thinlist}(Q)$	(8.6) $\text{preorder}(Q) \implies \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$
(h)	$\sqsubseteq \frac{F(\exists) ((f_1 \ p_1) \cup (f_2 \ p_2))}{\exists} \text{ thin}(Q) \text{ sort}(\leq)$	

§8.4. The knapsack problem

(8.4a)

	name	type	definition	meaning
1	<code>total(f,x)</code>	<code>Int</code>	$\text{total}(f,x) \triangleq \text{sum}(\text{list}(f,x))$, $\text{total } f = \text{list}(f) \text{ sum}$	add up f over the items of the list
2	<code>subseq</code>	<code>[A]→[A]</code>	$x \text{ subseq } ys \iff \text{subseq}(ys,x)$	every subsequence of the list
3	<code>[nil,cons]</code>	<code>F[Item]→[Item]</code>	$[\text{nil},\text{cons}] = [\text{nil},\text{cons}]$, $[\text{nil},\text{cons}](\text{l}(u)) = []$, $[\text{nil},\text{cons}](\text{r}((a,x))) = [a]\#x$	keep the item: the list constructor
4	<code>[nil,π_2]</code>	<code>Unit + Item × [Item]→[Item]</code>	$[\text{nil},\pi_2](\text{l}(D)) = []$, $[\text{nil},\pi_2](\text{r}(p)) = \pi_2(p)$, $[\text{nil},\pi_2] = [\text{nil},\pi_2]$	drop the item: keep the tail
5	<code>within(w)</code>	<code>[Item]→[Item]</code>	$\text{within}(w)(x,y) \iff x = y \wedge \text{total}(\text{wt},x) \leq w$	the packings whose total weight is at most w
6	$([\text{nil},\text{cons}]\text{within}(w)) \cup [\text{nil},\pi_2]$	<code>F[Item]→[Item]</code>	$([\text{nil},\text{cons}]\text{within}(w)) \cup [\text{nil},\pi_2] \triangleq ([\text{nil},\text{cons}]\text{within}(w)) \cup [\text{nil},\pi_2]$, $([\text{nil},\text{cons}]\text{within}(w)) \cup [\text{nil},\pi_2] = [\text{nil},(\text{cons within}(w)) \cup \pi_2]$, $([\text{nil},\text{cons}]\text{within}(w))(u,r) \iff r = [\text{nil},\text{cons}](u) \wedge \text{total}(\text{wt},r) \leq w$	at each item, keep it if the packing still fits, or drop it
7	<code>R</code>	<code>[Item]→[Item]</code>	$R = \text{total vol} \geq \text{total vol}^\circ$, $R(x,y) \iff \text{total}(\text{vol},y) \leq \text{total}(\text{vol},x)$	x is worth at least as much as y
8	<code>Q(vol,wt)</code>	<code>[Item]→[Item]</code>	$Q(\text{vol},\text{wt}) = R \cap (\text{total wt} \leq \text{total wt}^\circ)$, $Q(\text{vol},\text{wt})(x,y) \iff \text{total}(\text{vol},y) \leq \text{total}(\text{vol},x) \wedge \text{total}(\text{wt},x) \leq \text{total}(\text{wt},y)$	x is worth at least as much as y and weighs no more

(8.4b)

	the law	what it says
1	$(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$ FALSE	not monotonic on R: a selection of greater value need not still fit once one more item goes in
2	$F(Q(\text{vol},\text{wt})) [\text{nil},\text{cons}] \text{within}(w) \sqsubseteq [\text{nil},\text{cons}] \text{within}(w) Q(\text{vol},\text{wt})$ $F(Q(\text{vol},\text{wt})) [\text{nil},\pi_2] \sqsubseteq [\text{nil},\pi_2] Q(\text{vol},\text{wt})$,	both halves are monotonic on Q once ties in value are broken by weight
3	$F(R) [\text{nil},\text{cons}] \sqsubseteq [\text{nil},\text{cons}] R$ $F(R) [\text{nil},\pi_2] \sqsubseteq [\text{nil},\pi_2] R$	both algebras are monotonic on R itself, so R is the sort order P

(8.4a)

§8.5. The paragraph problem

(8.5a)

	name	type	definition	meaning
1	Line	Type	$\text{Line} \triangleq \text{L}(\text{Word})$	a line is a non-empty list of words
2	Para	Type	$\text{Para} \triangleq \text{L}(\text{Line})$	a paragraph is a non-empty list of lines
3	$\text{new}(a,p)$	Para	$\text{new}(a,p) \triangleq [[a]]\#p$	start a new first line holding only the word a
4	$\text{glue}(a)$	$\text{Para} \rightarrow \text{Para}$	$\text{glue}(a,[l]) = [[a]\#l], \text{glue}(a,[l]\#a') = [[a]\#l]\#a'$	put the word a at the front of the first line
5	$[\text{wrap wrap}, \text{new}]$	$\text{F}(\text{Para}) \rightarrow \text{Para}$	$[\text{wrap wrap}, \text{new}](l(a)) = [[a]], [\text{wrap wrap}, \text{new}](r(q)) = \text{new}(\pi_1(q), \pi_2(q))$	one word is a one-line paragraph; each further word starts a new line
6	$[\text{wrap wrap}, \text{glue}]$	$\text{F}(\text{Para}) \rightarrow \text{Para}$	$[\text{wrap wrap}, \text{glue}](l(a)) = [[a]], [\text{wrap wrap}, \text{glue}](r(q)) = \text{glue}(\pi_1(q), \pi_2(q))$	one word is a one-line paragraph; each further word joins the first line
7	$[\text{wrap wrap}, \text{new} \cup \text{glue}]$	$\text{F}(\text{Para}) \rightarrow \text{Para}$	$[\text{wrap wrap}, \text{new} \cup \text{glue}] \triangleq [\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}]$	each further word either starts a new line or joins the first line
8	partition	$\text{L}(\text{Word}) \rightarrow \text{Para}$	$\text{partition} \triangleq ([\text{wrap wrap}, \text{new} \cup \text{glue}]) ,$ $\text{partition}([a],q) \iff q = [[a]] ,$ $\text{partition}([a]\#x,q) \iff (\exists p. \text{partition}(x,p) \wedge (q = \text{new}(a,p) \vee q = \text{glue}(a,p)))$	every way of breaking the words into lines
9	width	$\text{Line} \rightarrow \text{Int}$	$\text{width}[a] = \text{len}(a), \text{width}([a]\#x) = (\text{len}(a) + \text{width}(x)) + 1$	the lengths of the words plus one space between neighbours
10	head	$\text{Para} \rightarrow \text{Line}$	$\text{head}[l] = l, \text{head}([l]\#a) = l$	the first line of the paragraph
11	$\text{fits}(w)$	$\text{P}(\text{Para})$	$\text{fits}(w,[l]) = \text{width}(l) \leq w, \text{fits}(w,[l]\#a) = \text{width}(l) \leq w \wedge \text{fits}(w,a)$	every line is at most w wide
12	$\text{fits}(w)$	$\text{Para} \rightarrow \text{Para}$	$\text{fits}(w)(p,q) \iff p = q \wedge \text{fits}(w,p)$	the paragraphs whose every line is at most w wide
13	$\text{ok}(w)$	$\text{Para} \rightarrow \text{Para}$	$\text{ok}(w)(p,q) \iff p = q \wedge \text{width}(\text{head}(p)) \leq w$	the paragraphs whose first line is at most w wide
14	$\text{sqr}(n)$	Int	$\text{sqr}(n) \triangleq nn$	n squared
15	$\text{waste}(w)$	$\text{Para} \rightarrow \text{Int}$	$\text{waste}(w,[l]) = 0, \text{waste}(w,[l]\#a) = \text{sqr}(w - \text{width } l) + \text{waste}(w,a)$	the squared white space left on every line but the last, added up
16	R	$\text{Para} \rightarrow \text{Para}$	$R = \text{waste } w \leq \text{waste } w^\circ, R(p,q) \iff \text{waste}(w,p) \leq \text{waste}(w,q)$	p wastes no more than q
17	$Q(\text{len},w)$	$\text{Para} \rightarrow \text{Para}$	$Q(\text{len},w) = R \cap (\text{head head}^\circ), Q(\text{len},w)(p,q) \iff \text{waste}(w,p) \leq \text{waste}(w,q) \wedge \text{head}(p) = \text{head}(q)$	p wastes no more than q and has the same first line
18	$\text{start}(a)$	[Para]	$\text{start}(a) \triangleq [[[a]]]$	the one candidate for a single word: the paragraph of one line holding that word

(8.5b)

the law	what it says
1 $(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$	glue is not monotonic on R : the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
2 $F(Q(\text{len}, w)) \text{ [wrap wrap, new]} \sqsubseteq \text{[wrap wrap, new]} Q(\text{len}, w)$ $(\forall a. 0 \leq \text{len}(a)) \Rightarrow F(Q(\text{len}, w)) \text{ [wrap wrap, glue]} \text{ ok}(w) \sqsubseteq \text{[wrap wrap, glue]} \text{ ok}(w) Q(\text{len}, w)$ $\text{cons monotonic on collect} \leq \text{collect}^\circ.$	both halves are monotonic on Q once ties in waste are broken by the first line
3 $\text{merge}(\tau) = \text{cat}$	τ needs no sorting at all
4 $F(\tau) \text{ [wrap wrap, new]} \sqsubseteq \text{[wrap wrap, new]} \tau$ $F(\tau) \text{ [wrap wrap, glue]} \sqsubseteq \text{[wrap wrap, glue]} \tau$	both algebras are monotonic on τ , so τ is the sort order P

$$(\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \Rightarrow (Q(\text{listcp } \langle g_1, g_2 \rangle \text{ merge}(\tau) \text{ thinlist}(Q(\text{len}, w)))) \text{ minlist}(R) \sqsubseteq \frac{\text{partition fits}(w)}{\tau} \text{ est}(R) \quad (8.5c)$$

a paragraph laid out as a fold that thins the layouts kept at each word

circuit	Hinze–Marsden
$\text{L}(\text{Word}) \xrightarrow{\frac{1}{\tau}} \text{P}(\text{L}(\text{Word})) \xrightarrow{\text{E}(\text{partition fits}(w))} \text{P}(\text{L}(\text{Line})) \xrightarrow{\text{est}(R)} \text{L}(\text{Line})$ $\frac{\text{partition L(fits } w)}{\tau} \text{ est}(R)$	
$\text{L}(\text{Word}) \xrightarrow{\frac{1}{\tau}} \text{P}(\text{L}(\text{Word})) \xrightarrow{\text{E}(\langle [\text{wrap wrap, new}] \cup \{[\text{wrap wrap, glue}] \text{ok}(w)\} \rangle)} \text{P}(\text{L}(\text{Line})) \xrightarrow{\text{est}(R)} \text{L}(\text{Line})}$ $= \frac{\langle [\text{wrap wrap, new}] \cup \{[\text{wrap wrap, glue}] \text{ok}(w)\} \rangle}{\tau} \text{ est}(R)$ fusion, every word fits on a line by itself.	
$\text{L}(\text{Word}) \xrightarrow{\frac{1}{\tau}} \text{P}(\text{L}(\text{Word})) \xrightarrow{\text{E}(\langle [\text{wrap wrap, new}] \cup \{[\text{wrap wrap, glue}] \text{ok}(w)\} \rangle)} \text{P}(\text{L}(\text{Line})) \xrightarrow{\text{est}(R)} \text{L}(\text{Line})}$ $= \frac{\langle [\text{wrap wrap, new}] \cup \{[\text{wrap wrap, glue}] \text{ok}(w)\} \rangle}{\tau} \text{ est}(R)$ the algebra as $(f_1 p_1) \cup (f_2 p_2)$, $p_1 \triangleq 1$ — binary thinning theorem.	
$\text{L}(\text{Word}) \xrightarrow{\text{F}(\text{list}(\text{L}(\text{Line})))} \text{listcp} \text{ list}(\text{F}(\text{L}(\text{Line}))) \xrightarrow{\begin{matrix} \text{0}_1 \\ \text{0}_2 \end{matrix}} \begin{matrix} \text{L}(\text{Line}) \\ \text{L}(\text{Line}) \end{matrix} \xrightarrow{\text{merge}(\tau)} \text{thinlist}(Q(\text{len}, w)) \xrightarrow{\text{list}(\text{L}(\text{Line}))} \text{minlist}(R) \xrightarrow{\text{L}(\text{Line})}$ $Q(\text{listcp } \langle g_1, g_2 \rangle \text{ cat thinlist}(Q)) \text{ minlist}(R)$ binary thinning theorem, at $P \triangleq T$ with $\text{merge } \tau = \text{cat}$ ($\text{merge}(\tau) = \text{cat}$), Q from $F(Q(\text{len}, w)) \text{ [wrap wrap, new]} \sqsubseteq \text{[wrap wrap, new]} Q(\text{len}, w)$, $(\forall a. 0 \leq \text{len}(a)) \Rightarrow F(Q(\text{len}, w)) \text{ [wrap wrap, glue]} \text{ ok}(w) \sqsubseteq \text{[wrap wrap, glue]} \text{ ok}(w) Q(\text{len}, w)$	
$Q[\text{start, cpr } \langle h_1, h_2 \rangle \text{ cat thinlist}(Q)] \text{ minlist}(R)$ $\text{listcp} = \text{wrap} + \text{cpr}$, g_i along the coproduct — $[\text{wrap wrap, new}]$ $= (\text{l}(a)) = [[a]], [\text{wrap wrap, new}](r(q)) = \text{new}(\pi_1(q), \pi_2(q)), [\text{wrap wrap, glue}](\text{l}(a)) = [[a]], [\text{wrap wrap, glue}](r(q)) = \text{glue}(\pi_1(q), \pi_2(q))$	

§8.6. Bitonic tours

(8.6a)

	name	type	definition	meaning
1	Journey(City)	Type	$\text{Journey}(\text{City}) \triangleq \mu F$	a list of at least two cities, the last two held as a pair
2	Tour(City)	Type	$\text{Tour}(\text{City}) \triangleq \text{Journey}(\text{City}) \times \text{Journey}(\text{City})$	the outward journey and the return journey
3	head	$\text{Journey}(\text{City}) \rightarrow \text{City}$	$\text{head}(\text{wrap}(p)) = \pi_1(p), \text{head}([a]\#a') = a$	the first city of the journey
4	next	$\text{Journey}(\text{City}) \rightarrow \text{City}$	$\text{next}(\text{wrap}(p)) = \pi_2(p), \text{next}([a]\#a') = \text{head}(a')$	the second city of the journey
5	$[a]\#\text{tail}(\cdot)$	$\text{Journey}(\text{City}) \rightarrow \text{Journey}(\text{City})$	$[a]\#\text{tail}(\text{wrap}(p)) = \text{wrap}(a, \pi_2(p)), [a]\#\text{tail}([a']\#a'') = [a]\#a''$	replace the first city of the journey by a
6	outcost	$\text{Journey}(\text{City}) \rightarrow \text{Int}$	$\text{outcost}(\text{wrap}(p)) = \text{tc}(p), \text{outcost}([a]\#a') = \text{tc}(a, \text{head}(a')) + \text{outcost}(a')$	the cost of travelling the journey forwards
7	incost	$\text{Journey}(\text{City}) \rightarrow \text{Int}$	$\text{incost}(\text{wrap}(p)) = \text{tc}(\pi_2(p), \pi_1(p)), \text{incost}([a]\#a') = \text{tc}(\text{head}(a'), a) + \text{incost}(a')$	the cost of travelling the journey backwards
8	cost(t)	Int	$\text{cost}(t) \triangleq \text{outcost}(\pi_1(t)) + \text{incost}(\pi_2(t))$	the outward journey forwards plus the return journey backwards
9	start	$\text{City} \times \text{City} \rightarrow \text{Tour}(\text{City})$	$\text{start}(a) \triangleq (\text{wrap}(a), \text{wrap}(a))$	the tour of the last two cities, both journeys being that one edge
10	dropl(a,t)	Tour(City)	$\text{dropl}(a,t) \triangleq ([a]\#\text{tail}(\pi_1(t)), [a]\#\pi_2(t))$	a replaces the first city of the outward journey and is put in front of the return
11	dropr(a,t)	Tour(City)	$\text{dropr}(a,t) \triangleq ([a]\#\pi_1(t), [a]\#\text{tail}(\pi_2(t)))$	a is put in front of the outward journey and replaces the first city of the return
12	[start,dropl]	$F(\text{Tour}(\text{City})) \rightarrow \text{Tour}(\text{City})$	$[\text{start}, \text{dropl}](l(p)) = \text{start}(p), [\text{start}, \text{dropl}](r(q)) = \text{dropl}(\pi_1(q), \pi_2(q))$	start at the last two cities, then add each city by dropl
13	[start,dropr]	$F(\text{Tour}(\text{City})) \rightarrow \text{Tour}(\text{City})$	$[\text{start}, \text{dropr}](l(p)) = \text{start}(p), [\text{start}, \text{dropr}](r(q)) = \text{dropr}(\pi_1(q), \pi_2(q))$	start at the last two cities, then add each city by dropr
14	[start,dropl u dropr]	$F(\text{Tour}(\text{City})) \rightarrow \text{Tour}(\text{City})$	$[\text{start}, \text{dropl} \cup \text{dropr}] \triangleq [\text{start}, \text{dropl}] \cup [\text{start}, \text{dropr}], [\text{start}, \text{dropl} \cup \text{dropr}](u, t) = t = [\text{start}, \text{dropl}](u) \vee t = [\text{start}, \text{dropr}](u)$	each further city is added by dropl or by dropr
15	$\llbracket [\text{start}, \text{dropl} \cup \text{dropr}] \rrbracket$	$\mu F \rightarrow \text{Tour}(\text{City})$	$\llbracket [\text{start}, \text{dropl} \cup \text{dropr}] \rrbracket \triangleq \llbracket [\text{start}, \text{dropl} \cup \text{dropr}] \rrbracket$	every bitonic tour of the cities
16	R	$\text{Tour}(\text{City}) \rightarrow \text{Tour}(\text{City})$	$R = \text{cost} \leq \text{cost}^\circ, R(t, t') \iff \text{cost}(t) \leq \text{cost}(t')$	t costs no more than t'
17	next2(t)	$\text{City} \times \text{City}$	$\text{next2}(t) \triangleq (\text{next}(\pi_1(t)), \text{next}(\pi_2(t)))$	the second cities of both journeys
18	head2(t)	$\text{City} \times \text{City}$	$\text{head2}(t) \triangleq (\text{head}(\pi_1(t)), \text{head}(\pi_2(t)))$	the first cities of both journeys
19	Q	$\text{Tour}(\text{City}) \rightarrow \text{Tour}(\text{City})$	$Q = R \cap (\text{next2} \text{ next2}^\circ) \cap (\text{head2} \text{ head2}^\circ), Q(t, t') \iff \text{cost}(t) \leq \text{cost}(t') \wedge (\text{next}(\pi_1(t)) = \text{next}(\pi_1(t')) \wedge \text{next}(\pi_2(t)) = \text{next}(\pi_2(t'))) \wedge (\text{head}(\pi_1(t)) = \text{head}(\pi_1(t')) \wedge \text{head}(\pi_2(t)) = \text{head}(\pi_2(t')))$	t costs no more than t', and both journeys of t and t' share their first two cities

(8.6b)

the law	what it says
1 $(1 \times R) \text{ drop} \sqsubseteq \text{drop} \text{ } R \text{ } \text{FALSE}$ $(1 \times R) \text{ drop} \sqsubseteq \text{drop} \text{ } R \text{ } \text{FALSE}$	neither drop is monotonic on R : the two edges it adds and removes depend on head and next of both lists
2 $F(Q) \text{ } [start, \text{drop} \text{ } l] \sqsubseteq [start, \text{drop} \text{ } l] \text{ } Q$ $F(Q) \text{ } [start, \text{drop} \text{ } r] \sqsubseteq [start, \text{drop} \text{ } r] \text{ } Q$,	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input
3 $F(\tau) \text{ } [start, \text{drop} \text{ } l] \sqsubseteq [start, \text{drop} \text{ } l] \text{ } \tau$ $F(\tau) \text{ } [start, \text{drop} \text{ } r] \sqsubseteq [start, \text{drop} \text{ } r] \text{ } \tau$	both algebras are monotonic on τ , so τ is the sort order P

$$(\text{listcp } (g_1, g_2) \text{ cat thinlist}(Q)) \text{ minlist}(R) \sqsubseteq \frac{([start, \text{drop} \text{ } l \text{ } u \text{ } \text{drop} \text{ } r])}{\tau} \text{ est}(R) \quad (8.6c)$$

a least-cost bitonic tour, as a fold that thins the tours kept at each city

circuit	Hinze–Marsden
$\mu F \text{ } \left[\frac{1}{\tau} \right] \text{ } P(\mu F) \text{ } \text{E}([start, \text{drop} \text{ } l \text{ } u \text{ } \text{drop} \text{ } r]) \text{ } P(\text{Journey}(\text{City})^2) \text{ } \text{est}(R) \text{ } \text{Journey}(\text{City})$ $\frac{\text{tour}}{\tau} \text{ est}(R)$	
$\mu F \text{ } \left[\frac{1}{\tau} \right] \text{ } P(\mu F) \text{ } \text{E}([start, \text{drop} \text{ } l \text{ } u \text{ } [start, \text{drop} \text{ } r]]) \text{ } P(\text{Journey}(\text{City})^2) \text{ } \text{est}(R) \text{ } \text{Journey}(\text{City})$ $= \frac{([start, \text{drop} \text{ } l \text{ } u \text{ } \text{drop} \text{ } r])}{\tau} \text{ est}(R)$ $([start, \text{drop} \text{ } l \text{ } u \text{ } \text{drop} \text{ } r]) \triangleq ([start, \text{drop} \text{ } l \text{ } u \text{ } \text{drop} \text{ } r])$	
$\mu F \text{ } \left[\frac{1}{\tau} \right] \text{ } P(\mu F) \text{ } \text{E}([start, \text{drop} \text{ } l \text{ } u \text{ } \text{drop} \text{ } r]) \text{ } P(\text{Journey}(\text{City})^2) \text{ } \text{est}(R) \text{ } \text{Journey}(\text{City})$ $= \frac{([start, \text{drop} \text{ } l \text{ } u \text{ } \text{drop} \text{ } r])}{\tau} \text{ est}(R)$ $([start, \text{drop} \text{ } l \text{ } u \text{ } \text{drop} \text{ } r]) \triangleq ([start, \text{drop} \text{ } l \text{ } u \text{ } \text{drop} \text{ } r])$ $(\text{listcp } (g_1, g_2) \text{ cat thinlist}(Q)) \text{ minlist}(R)$ binary thinning theorem, at $P \triangle T$ with merge $\tau = \text{cat}$, Q from $F(Q)$ $[start, \text{drop} \text{ } l] \sqsubseteq [start, \text{drop} \text{ } l] \text{ } Q, F(Q) \text{ } [start, \text{drop} \text{ } r] \sqsubseteq [start, \text{drop} \text{ } r] \text{ } Q$	
$([start \text{ wrap, cpr } (list(\text{drop} \text{ } l), list(\text{drop} \text{ } r)) \text{ cat thinlist}(Q)) \text{ minlist}(R)$ $listcp = \text{wrap} + \text{cpr}, g_i = [list(start), list(drop_i)] \text{ } \text{---}$ $= [start, \text{drop} \text{ } l](l(p)) = start(p), [start, \text{drop} \text{ } l](r(q)) = \text{drop} \text{ } l(\pi_1(q), \pi_2(q)), [start, \text{drop} \text{ } r](l(p)) = start(p), [start, \text{drop} \text{ } r](r(q)) = \text{drop} \text{ } r(\pi_1(q), \pi_2(q)); \text{quadratic, two tours added per step}$	